

DATA ANALYSIS PORTFOLIO PROJECT

Northwind Sales Analysis

A Business Intelligence Report

A comprehensive analysis of Northwind Traders' sales, customer behaviour, supplier distribution, and operational performance combining SQL querying, Power BI dashboarding, and Excel scenario modelling.

\$1.27M TOTAL REVENUE	2,155 TOTAL ORDERS	91 CUSTOMERS	77 PRODUCTS
---	--	------------------------------------	-----------------------------------

Prepared by: Samarpit
Date: May 2026
Dataset: Northwind Traders (MySQL)

14+ SQL Queries | 5 Power BI Dashboards
| 3 Excel Models
Tools: MySQL | Power BI | Excel

EXECUTIVE SUMMARY

What This Analysis Found

Northwind Traders generated \$1.27M in total revenue across 2155 orders from 91 customers worldwide. This report identifies revenue concentration risks, seasonal order patterns, shipping inefficiencies, and customer segmentation opportunities.

Revenue Leader: Beverages

- 1 Beverages is the #1 category contributing ~21% of total revenue (\$267,868). However, a single product — Cote de Blaye at \$141,397 — accounts for the majority of that figure, creating concentration risk.

Q3 Order Volume Collapse

- 2 Order volume dropped sharply from ~15,000 units in Q1-Q2 to ~8,000 in Q3 — a 47% fall. This seasonal dip represents a significant untapped revenue opportunity requiring a targeted promotional strategy.

Shipping Punctuality: Only 36%

- 3 Fewer than 4 in 10 orders arrive on time. Average shipping takes 8.49 days with a maximum of 37 days. Ireland, Sweden and Switzerland are the worst-performing regions — a critical operational risk.

KEY STRENGTHS

- **Active Customer Rate:** 73 of 91 customers placed orders — an 80% retention rate
- **High-Value Accounts:** QUICK, SAVEA, and ERNSH are top-tier customers with \$100K+ combined spend
- **Category Diversity:** 8 product categories provide revenue spread across multiple lines

KEY RISKS

- **Revenue Concentration:** One SKU (Cote de Blaye) drives disproportionate Beverages revenue
- **Logistics Gap:** 64% of shipments are late — direct customer satisfaction risk
- **Supplier Cost:** France avg unit price \$76.75 vs global avg ~\$21 — margin pressure

PROJECT CONTEXT

About This Project

WHAT IS NORTHWIND?

- **Dataset:** Northwind Traders is a standard sample business database simulating a real trading company with customers, suppliers, employees, products and orders across multiple countries
- **Purpose:** End-to-end BI practice covering all major analyst workflows, SQL querying, dashboard design, and Excel scenario modelling

DATA SCOPE

- **8 Product Categories**, 77 Products, 91 Customers
- **29 Suppliers**, 9 Employees, 21 Countries
- **2,155 Orders**, 14 SQL Queries, 5 Dashboards

\$1.27M TOTAL REVENUE	\$1.53K AVG ORDER VALUE	\$14.22K AVG CUSTOMER LTV	61.83 AVG ORDER QTY
---------------------------------	-----------------------------------	-------------------------------------	-------------------------------

TOOLS & METHODS

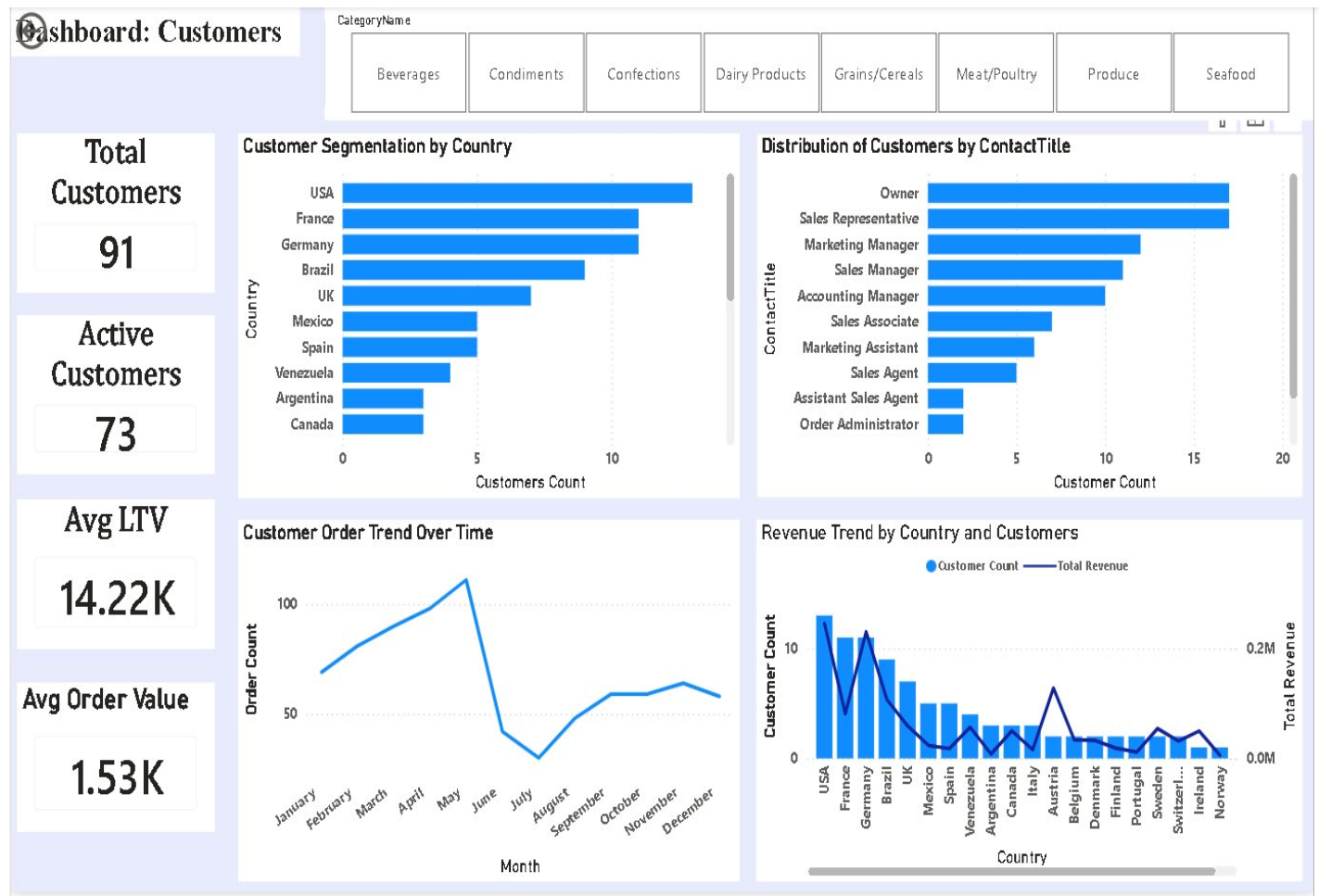
- **MySQL:** 14 queries — JOINS, CTEs, window functions (LAG), correlated subqueries, HAVING
- **Power BI:** 5 interactive dashboards: Customers, Sales, Operations, Suppliers, Employees
- **Excel:** Goal Seek scenario model, customer segmentation, supplier pricing flags

ANALYSIS APPROACH

- **Question-first:** Each SQL query mapped to a specific business question
- **Cross-domain:** Findings connect customers, products, operations and suppliers
- **Actionable:** Every insight paired with a business recommendation

Customer Analysis

POWER BI — DASHBOARD: CUSTOMERS



- **Geographic Concentration:** USA leads in customer count, followed by France and Germany. These 3 countries form the core of Northwind's customer geography.
- **Active Customer Rate:** 73 of 91 customers placed orders (80% active rate). The 18 inactive customers represent a re-engagement opportunity worth ~\$25K+ in potential revenue at average LTV.
- **Seasonal Peak in May:** Order volume peaks in May (~115 orders) then collapses to July lows (~30 orders), suggesting strong seasonal effects or promotional dependency.
- **Customer Profile:** Top contact titles are Owner and Sales Representative — Northwind primarily serves small-to-mid businesses where decision-makers place orders directly.

Sales & Product Analysis

POWER BI — DASHBOARD: SALES

Dashboard: Sales



- **Cote de Blaye Outlier:** At \$141,397 in revenue, Cote de Blaye outperforms the #2 product (Thuringer Rostbratwurst ~\$80K) by 76%. Single-SKU dependency is a strategic supply risk.
- **Category Revenue Distribution:** The treemap confirms Beverages as the largest revenue block, with Dairy Products and Confections as 2nd and 3rd. Produce and Grains/Cereals are smallest.
- **Q3 Order Volume Drop:** Volume drops to ~8K units in Q3 from a Q1 high of ~16K. A targeted mid-year promotion could partially recover this 47% gap.
- **Beverage Discount Pressure:** The Discount vs Order Count chart shows Beverages receives the most discounting — high order count but compressed margins.

DASHBOARD 3 OF 5

Operations & Shipping

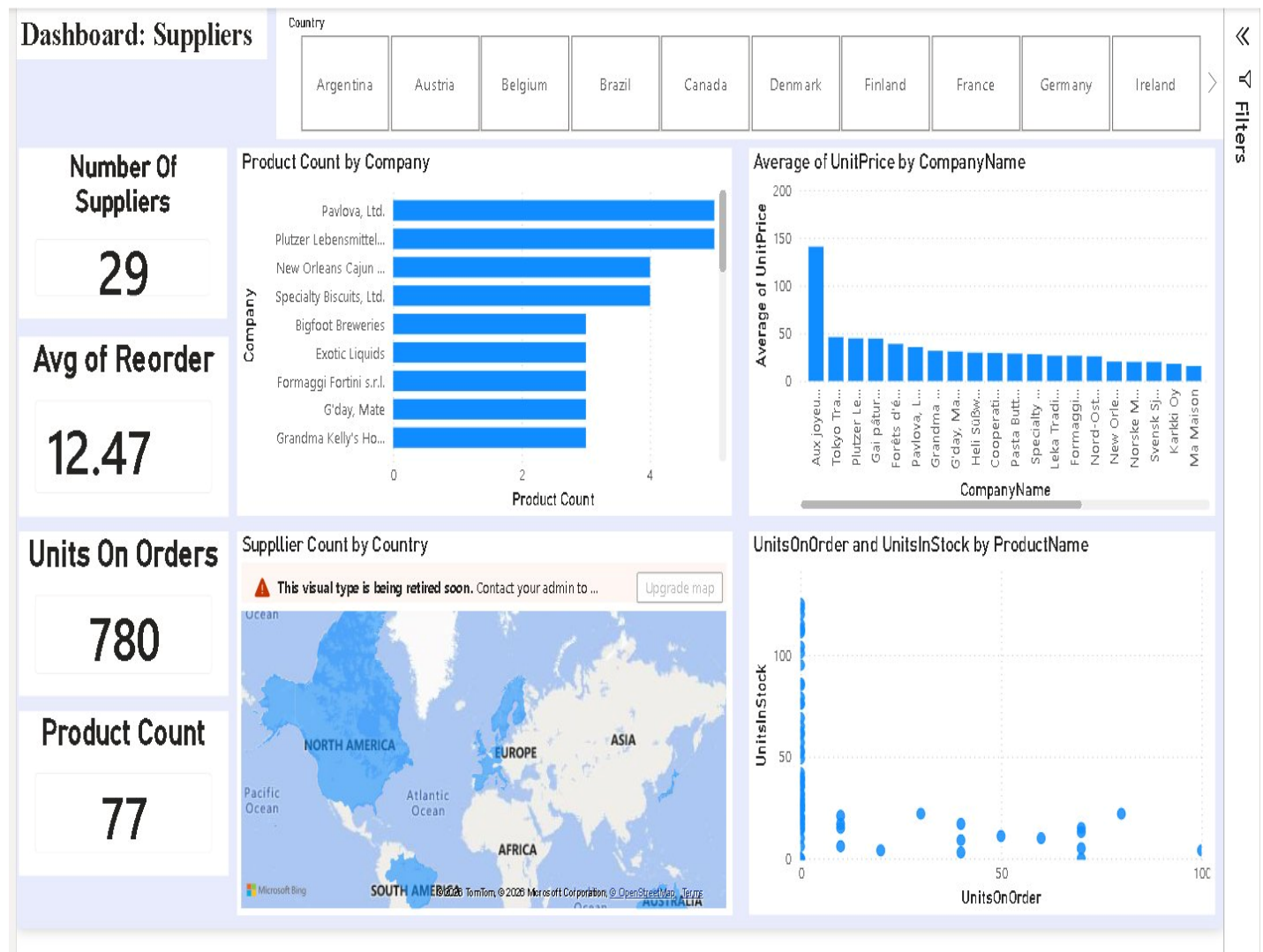
POWER BI — DASHBOARD: OPERATIONS



- **36% Punctuality — Critical Issue:** Fewer than 4 in 10 orders arrive on time. This is the single biggest operational red flag in the entire dataset and directly impacts customer retention.
- **Ireland Worst Performing Region:** Ireland tops the delayed shipment list at ~12 days average, followed by Sweden and Switzerland. These three markets require immediate logistics review.
- **Max 37-Day Delay:** While the average is 8.49 days, a 37-day maximum indicates extreme outlier cases needing root cause analysis — likely tied to specific carriers or routes.
- **Seafood Overstocked:** Seafood has ~700 units in stock — the highest of all categories — yet is not a top revenue contributor. Likely overstocking and tied-up working capital.

Supplier Analysis

POWER BI — DASHBOARD: SUPPLIERS



- **USA: Most Suppliers, Lowest Price:** USA has 12 suppliers at avg \$20.87 unit price — the most cost-efficient sourcing. Prioritising US suppliers for high-volume categories can improve margins.
- **France: Expensive Outlier:** France has only 5 suppliers but avg unit price of \$76.75 — nearly 4x the global average. Unless products command premium customer pricing, this is a margin risk.
- **Pavlova Ltd. Concentration Risk:** Pavlova Ltd. has the highest product count per company — a supplier concentration risk if one company supplies multiple critical SKUs.
- **Stock-Order Imbalance:** Some products show high on-order quantities but low stock levels — potential supply chain delays affecting fulfilment in specific product lines.

DASHBOARD 5 OF 5

Employee Analysis

POWER BI — DASHBOARD: EMPLOYEES



- **Top Revenue Performers:** Margaret Peacock is the top revenue-generating employee at ~\$0.25M, followed closely by Janet Leverling and Nancy Davolio. Top 3 account for ~60% of employee-attributed revenue.
- **Uniform Tenure Window:** All 9 employees have 32-34 years of tenure — an experienced team but also a succession planning risk with multiple potential retirements in the same window.
- **Workforce Profile:** Anne Dodsworth is the most recent joiner. Sales Representatives dominate the title distribution, as expected for a trading company.
- **Hiring Pattern:** Hiring is evenly distributed across all 4 quarters historically — no seasonal hiring pattern visible in the data.

STRATEGIC RECOMMENDATIONS

What Should Be Done

Based on findings across all five analysis domains, the following actions are recommended in order of business impact:

#	Area	Problem Identified	Recommendation	Priority
1	Operations	Only 36% of orders ship on time. Ireland, Sweden, Switzerland are worst performers.	Audit logistics partners for top 3 delayed countries. Set SLA targets per carrier. Track monthly.	High
2	Revenue Risk	Cote de Blaye (single SKU) accounts for a disproportionate share of Beverages revenue at \$141K.	Diversify by actively promoting 2-3 underperforming Beverages SKUs to reduce single-product dependency.	High
3	Seasonality	Q3 order volume drops 47% from Q1 peak. July is the lowest single month.	Launch mid-year promotion or loyalty incentive in June-August targeting the top 20 customers by spend.	Medium
4	Customers	18 customers are inactive (20% of total base). Average LTV is \$14.22K.	Run re-engagement campaign for inactive customers. 5 conversions = ~\$71K recovered revenue.	Medium
5	Supplier Cost	France supplier avg price (\$76.75) is 3.7x the global average of ~\$21.	Review French supplier contracts. Substitute with US or German alternatives for non-premium SKUs.	Low
6	Inventory	Seafood has highest stock (~700 units) but is not a top revenue category.	Reduce Seafood reorder quantities. Redirect freed working capital toward Meat/Poultry (high revenue, lower stock).	Low

TECHNICAL APPENDIX

Methodology & SQL Queries

TECH STACK

- **MySQL:** 14+ queries using advanced SQL — CTEs, window functions (LAG, PARTITION BY), correlated subqueries, HAVING
- **Power BI:** 5 dashboards with KPI cards, bar/line/treemap/scatter visuals, and country-level filters
- **Excel:** Goal Seek model, SUMIF/COUNTIF customer segmentation, IF-based supplier pricing flags with dynamic threshold

ANALYSIS DOMAINS

- **Customers (Q1-Q5):** Order frequency, geography, segmentation, purchase interval analysis using LAG()
- **Products & Revenue (Q4, Q9-Q11):** Category revenue, pricing vs stock, demand seasonality, anomaly detection
- **Employees (Q6-Q8):** Geographic distribution, hire trends, title patterns
- **Suppliers (Q12-Q14):** Regional pricing, category distribution, correlated subquery for top-priced category per country

ALL 14 SQL QUERIES

Q1 Average Orders per Customer Identifies high-value repeat customers; customers with 15+ orders flagged	Q2 Order Patterns by City & Country Geographic distribution of order volume across cities and countries
Q3 Customer Spend by Category Multi-table JOIN to cluster customers by category preference and total spend	Q4 Category & Product Revenue Top revenue contributors; category-country order correlation analysis
Q5 Order Frequency Analysis Uses LAG() window function to calculate avg days between orders per customer	Q6 Employee Geographic Distribution Title and country breakdown of the 9-person employee headcount
Q7 Hire Date Trends Annual hiring patterns over time using YEAR() function	Q8 Title & Courtesy Title Patterns Distribution of employee titles and courtesy titles by count
Q9 Price, Stock & Sales Correlation Compares unit price and stock levels against actual sales performance	Q10 Seasonal Demand by Category Quarterly order and unit sales volume per category (Q10 data)
Q11 Revenue Anomaly Detection Identifies bottom 5 underperforming products by revenue	Q12 Supplier Regional Pricing Country-level supplier count and average unit price comparison
Q13 Supplier-Category Distribution Maps supplier companies to product categories by count	Q14 Highest-Priced Category per Country Correlated subquery to find top avg-priced category per supplier country